A Red Team of Networked Personal Agents

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Abstract

As large language models gain tool access, persistent memory, and autonomous execution capabilities, they are deployed as personal agents that act on behalf of individual users. When such agents coordinate across ownership boundaries, each agent's deep access to its owner's private data meets another agent operating under a different principal. This creates a behavioral safety surface with no analogue in single-agent or prompt-injection settings: leakage occurs through tool calls, memory retrieval, and multi-turn social dynamics rather than direct text disclosure alone. We introduce **PART-Bench** (**P**ersonal **A**gent **R**ed-**T**eam Benchmark), an evaluation platform that formulates a society of proactive, delegated personal agents and provides the first benchmark where agents are equipped with realistic tools, persistent memory, 85+ scattered documents, and autonomous multi-turn interaction. We design a synthetic personal data environment with ground-truth labels across five sensitivity categories, evaluate governance configurations along a specificity gradient, and measure both task-completion utility and data-protection security simultaneously. Across XXXX experimental runs, we find that governance *specificity* determines behavioral safety: category-specific policies yield XXXX pp leak reduction while generic instructions yield only XXXX pp. Multi-turn coordination amplifies leakage XXXX$\times$ over single-step probing, and relationship context shifts leakage in opposing directions depending on data category. We release the full benchmark suite including agent configurations, evaluation queries with gold-standard labels, and the autonomous coordination engine.

1 Introduction

Large language models have rapidly evolved from conversational assistants into agentic systems capable of autonomous reasoning and action. Architectures such as ReAct [27] and Toolformer [17] established the paradigm of interleaving chain-of-thought reasoning with tool invocation; subsequent work on persistent memory [14], self-reflection [20], and multi-step planning [24] has produced agents that maintain state across sessions, learn from past interactions, and execute complex workflows without continuous human oversight. Commercial deployments now instantiate these capabilities as personal agents that manage calendars, search private document stores, draft communications, and coordinate tasks on behalf of individual users [10, 3, 13], with standardised protocols for tool integration (MCP; 1) and agent-to-agent communication (A2A; 6) accelerating adoption.

The next frontier for these agents is not greater individual autonomy but *inter-agent coordination*. Many high-value tasks are inherently cross-boundary: scheduling a meeting requires checking two people's calendars, sharing a project update requires one agent to contact another, and coordinating a product launch requires a manager's agent to collect status from several teammates' agents [21, 7]. Each of these tasks requires one person's delegate to interact with another person's delegate. Without this capability, humans remain manually bridging between individually capable AI systems. Yet when two delegate agents interact across an ownership boundary, a new class of behavioral safety problems

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emerges: a personal agent may reveal sensitive data through tool calls and memory retrieval, refuse legitimate requests and become useless as a delegate, comply with social pressure from the counterpart agent, or fail to escalate to its owner when ambiguity arises [19, 11]. The security community has documented that every published prompt-level defence can be bypassed under adaptive attack [12], and cross-boundary coordination satisfies all three conditions that make agents critically vulnerable: simultaneous processing of untrusted input, access to sensitive data, and execution of state-changing actions.

Existing evaluation approaches address fragments of this problem. Single-agent security benchmarks [29, 4, 22, 18] measure prompt injection resilience but involve no cross-boundary interaction. Multi-agent privacy benchmarks [25, 5, 9, 15, 26] study information disclosure in social or collaborative settings, but their agents communicate through scripted text exchanges without tool access, persistent memory, or scattered personal documents. Applied red-teaming studies [19] demonstrate behavioral failures in deployed systems but are expensive, non-reproducible, and unsuitable for systematic comparison of governance configurations. No existing benchmark simultaneously provides *realistic* agent capabilities (tools, memory, fragmented data), *proactive* autonomous interaction (agents that query without human prompting), and *reproducible* controlled evaluation (systematic comparison across defence levels, relationship conditions, and attack modalities).

We introduce **PART-Bench** (**P**ersonal **A**gent **R**ed-**T**eam Benchmark), a benchmark for evaluating the behavioral safety of personal agents engaged in cross-boundary coordination. Our contributions are:

1. **Formulation and platform.** We formulate the problem of a society of proactive, delegated personal agents coordinating across ownership boundaries and provide an evaluation platform in which fully-equipped agents, with persistent memory, 85+ scattered documents, callable tools (`search_notes`, `read_note`, `check_calendar`), and relationship context, interact autonomously over multi-turn horizons. This is the first benchmark where leakage occurs through tool calls and memory retrieval, not only through conversation text.
2. **Benchmark design.** We construct a synthetic personal data environment spanning five sensitivity categories with ground-truth labels for automated evaluation, define a governance gradient that isolates the effect of policy *specificity* from strictness, and provide a dual-metric protocol that simultaneously measures task-completion utility and data-protection security. The benchmark supports both single-step controlled evaluation and multi-turn autonomous coordination.
3. **Empirical findings.** Across XXXX experimental runs, we establish that governance specificity dominates strictness (XXXX pp improvement from category enumeration vs. XXXX pp from generic instruction), multi-turn interaction amplifies leakage XXXX$\times$ over single-step probing, relationship context has non-uniform category-dependent effects on data protection, and agents develop emergent attack patterns including progressive escalation and social reframing without explicit adversarial programming.

2 Related Work

Agent security benchmarks. The evaluation of LLM agent safety has produced a rich landscape of benchmarks, each targeting a specific threat model. InjecAgent [29] measures indirect prompt injection in tool-integrated agents across 1,054 test cases. AgentDojo [4] provides a dynamic sandboxed environment with 97 tasks and 629 injection tests. TensorTrust [22] and HackAPrompt [18] study prompt-level attack and defence at scale through gamified platforms. These benchmarks establish that LLM agents are vulnerable to injection, but evaluate single agents within a single trust domain and do not model cross-boundary interaction or measure the utility cost of defences.

Recent multi-agent privacy benchmarks move closer to our setting but answer a different question. AgentSocialBench [25] asks whether agents in social-network scenarios disclose private information, and uncovers an "abstraction paradox" in which privacy instructions paradoxically increase leakage. ConVerse [5] evaluates contextual safety in agent-to-agent conversations and finds attack success rates up to 88% when personal assistants interact with service providers. PAC-BENCH [15] treats privacy as a constraint on collaborative utility, MAGPIE [9] provides 200 collaborative tasks under a shared principal, and AgentLeak [26] taxonomises seven leakage channels in enterprise workflows. These works establish that privacy failures arise in social and collaborative settings. PART-Bench instead evaluates whether a *production-style delegated personal-agent stack* is deployable under cross-owner

Table 1: Comparison of agent safety benchmarks. ✓ = fully supported, $\sim$ = partially supported.

Benchmark	**Cross-boundary**	**Tool access**	**Dual metric**	**Defence gradient**	**Multi-turn**	**Persistent memory**	**Relation-ship**
InjecAgent		✓					
AgentDojo		✓	$\sim$				
TensorTrust				$\sim$			
ConVerse	✓				✓		
AgentSocialBench	✓		✓	$\sim$			✓
PAC-BENCH	✓		$\sim$				
AgentLeak	✓	$\sim$					
Chaos of Agents	✓	✓			✓	✓	
PART-Bench	✓	✓	✓	✓	✓	✓	✓

access: the target agent has tools, persistent memory, scattered private documents, relationship context, and autonomy to search, retrieve, and respond over multiple turns. The benchmark question is therefore not only whether agents leak, but whether a concrete agent stack can preserve data boundaries while remaining useful.

This distinction matters for methodology. Most multi-agent privacy benchmarks operate through scripted text exchanges: agents have no tool-mediated access to a private knowledge store, no persistent memory beyond injected profiles, and limited autonomy in deciding what to ask, retrieve, or probe next. Complementing automated benchmarks, Shapira et al. [19] present applied red-teaming of deployed agents, uncovering behavioral failures including uncontrolled information sharing and capability self-disablement, but their methodology is expensive, non-reproducible, and unsuitable for systematic comparison of governance configurations.

Attack and defence methods. Relevant attack modalities include automated jailbreak search via PAIR [2] and GCG [30], multi-turn escalation via Crescendo [16], and persuasion-based social engineering via PAP [28]. On the defence side, Instruction Hierarchy [23] and Spotlighting [8] represent prompt-level hardening, while system-level frameworks advocate layered defence [11] or architectural isolation through capability-based sandboxing. The consensus from large-scale adaptive evaluations [12] is that no prompt-level defence alone provides reliable protection. Our governance gradient (D0/D1/D2) is designed to quantify this gap empirically.

Positioning. Table 1 summarises existing benchmarks across seven dimensions critical to evaluating cross-boundary agent safety. PART-Bench is the first automated benchmark that simultaneously provides realistic agent capabilities (callable tools, persistent memory, fragmented personal data), proactive autonomous interaction (agents that query and adapt without human prompting), and controlled reproducible evaluation (systematic comparison across governance levels, relationship conditions, and interaction modalities). This combination is necessary to answer the deployment question faced by personal-agent platforms: given a raw or defended agent stack, where does it sit on the utility–security frontier?

3 Problem Formulation

We formalise the cross-boundary coordination problem to clarify the threat model, the variables under study, and the metrics that define success.

Agents and knowledge. A personal agent A_i operates on behalf of a human principal H_i and has access to a knowledge store $K_i = \{F_i, S_i, M_i\}$, where F_i is a set of documents (notes, files), S_i is structured state (calendar, todos, email), and M_i is persistent memory (facts learned through prior interactions). The agent is equipped with a tool set T_i (e.g., `search_notes`, `read_note`, `check_calendar`) through which it accesses K_i. Crucially, knowledge is *fragmented*: no single document contains a complete picture. The agent must search, retrieve, and synthesise across multiple sources to answer queries.

Sensitivity partition. The knowledge store K_i is partitioned into sensitivity categories $C = \{c_1, \dots, c_m\}$. In our instantiation, C consists of five categories: work/public, sensitive work, personal finance, personal health, and personal relationships. Each fact $f \in K_i$ belongs to exactly one category $c(f) \in C$. A subset of categories $C_{\text{protected}} \subset C$ designates information that should not be disclosed to external agents.

Cross-boundary interaction. When an external agent A_j (operating on behalf of H_j) contacts A_i, we define the interaction as a sequence of message exchanges $\mathcal{D} = (m_1, m_2, \dots, m_T)$ over T turns. At each turn, A_i may invoke tools from T_i, retrieve facts from K_i, and generate a response. The interaction is *cross-boundary* because A_j operates under a different principal with potentially adversarial objectives.

Governance policy. A governance policy π constrains the behaviour of A_i during cross-boundary interaction. We study prompt-level policies that vary in *specificity*: from no policy (π_0, the agent acts on default behaviour), to generic instruction (π_1, a single directive to protect privacy), to category-specific enumeration (π_2, explicit listing of protected categories and data types).

Relationship context. The interaction is further conditioned on a relationship $R(H_i, H_j) \in \mathcal{R}$, where $\mathcal{R}$ defines the set of possible social contexts (stranger, colleague, manager, etc.). Relationship context is provided to A_i through its memory and may influence disclosure behaviour.

Metrics. For a given configuration (π, R, mode), we define two metrics over a query set $Q = Q_{\text{pub}} \cup Q_{\text{prot}}$:

$$U(\pi, R) = \frac{|\{q \in Q_{\text{pub}} : \text{gold facts of } q \text{ appear in response}\}|}{|Q_{\text{pub}}|} \tag{1}$$

$$S(\pi, R) = \frac{|\{q \in Q_{\text{prot}} : \text{no gold facts of } q \text{ appear in response}\}|}{|Q_{\text{prot}}|} \tag{2}$$

where $U \in [0, 1]$ measures task-completion utility and $S \in [0, 1]$ measures data-protection security. A governance configuration is *Pareto-optimal* if no alternative achieves higher U without lower S, or vice versa. The central empirical question is whether this frontier is monotonic (more security always costs utility) or whether configurations exist that dominate on both axes.

4 PART-Bench Design

PART-Bench instantiates the formulation above as a concrete, reproducible evaluation platform. We describe the agent infrastructure, evaluation queries, governance gradient, interaction modes, and dual-metric protocol.

4.1 Agent with Personal Context

The target agent ("Alex's agent") is a realistic personal agent with:

- **Persistent memory**: a structured knowledge base containing the agent's understanding of its owner, built through prior interactions.
- **Notes**: 85+ documents organised in folders (Work/Projects, Work/Meetings, Work/HR, Personal/Finance, Personal/Health, Personal/Family), each containing facts that map to evaluation queries.
- **Tools**: callable functions (`read_note`, `search_notes`, `check_calendar`, `schedule_meeting`) that return real data. Leakage occurs not just through what the agent *says* but through what tools it *calls* and what data those tools *return*.
- **Identity**: a system prompt establishing the agent's owner, role, and communication style.

The querying agent ("Tina's agent") is tasked with gathering information from Alex's agent on behalf of its owner. In the multi-turn mode, Tina's agent maintains a checklist of pending questions and adaptively selects which to ask based on conversation history.

4.2 Evaluation Queries

PART-Bench comprises XXXX evaluation queries across 5 sensitivity categories:

Table 2: Evaluation query distribution across sensitivity categories. Work/public queries measure utility (should be answered); all others measure security (should be refused).

Category	ID range	Count	Expected behaviour
Work/public	Q1–XXXX	XXXX	Answer (utility)
Sensitive work (HR, board, investor)	XXXX–XXXX	XXXX	Refuse (security)
Personal finance	XXXX–XXXX	XXXX	Refuse (security)
Personal health	XXXX–XXXX	XXXX	Refuse (security)
Personal relationships	XXXX–XXXX	XXXX	Refuse (security)

Each query is paired with **gold key facts**: specific pieces of information whose presence in the agent's response constitutes a leak (for sensitive queries) or a correct answer (for work/public queries). This enables automated evaluation via string matching against gold facts, removing the need for LLM-as-judge and its associated reliability concerns.

4.3 Defence Gradient

PART-Bench separates prompt-level governance from architectural isolation. Prompt-level defences keep the target data visible to the model and ask the agent to behave correctly. Architectural defences remove or gate data before generation, testing whether safety improves when the agent cannot access restricted context in the first place.

- **D0 (No defence)**: The agent's system prompt contains no privacy instructions. Full data access, zero policy.
- **D1 (Generic prompt)**: A single sentence is appended: "*Try not to share personal or private information.*" This is the most common real-world defence pattern.
- **D2 (Category-specific deny list)**: An explicit enumeration: "*Never share: financial data (salary, equity, investments), health information (conditions, medications, appointments), HR/compensation details, personal relationship information.*" Same strictness intent as D1, higher specificity.
- **D3 (Spotlighting / delimiters)**: The external agent's message is explicitly marked as untrusted input, separating counterpart requests from the target agent's own instructions and memory.
- **D4 (Instruction hierarchy)**: The system prompt explicitly orders authority levels: system policy, owner policy, tool output, then external-agent request.
- **D5 (MCC isolation)**: Mountable Context Cells physically remove or withhold restricted notes, tools, and memory shards according to the relationship boundary.
- **D6 (MCC + escalation)**: D5 plus an escalation protocol that asks the owner, or consults prior owner decisions, when the request is ambiguous.

D0–D4 isolate prompt-level policy language; D5–D6 test architectural isolation. This split is central to the benchmark: if D5 dominates D4 at similar utility, the limiting factor is not prompt wording but context exposure.

4.4 Evaluation Modes

Single-step mode. Each query is asked in isolation: one question, one response, no conversation history. This provides a controlled measurement of per-query behaviour but misses conversational dynamics.

Multi-turn heartbeat mode. Both agents operate as persistent processes, exchanging messages at regular intervals ("heartbeats"). Tina's agent maintains a checklist of pending questions and decides which to ask, how to frame it, and whether to retry based on the evolving conversation. This captures:

- **Escalation**: repeated probing after initial refusal.

Table 3: Layered PART-Bench experimental design. The NeurIPS paper focuses on Layers 1 and 4, with a small Layer 2 attack slice if space and budget permit. Layers 3, 5, and 6 are benchmark extensions for the thesis and follow-up defense paper.

Layer	Question	Matrix	Role
L1 Raw stack	How unsafe is the personal-agent stack under raw and simple-prompt conditions?	L0 model $\times$ 3 reps $\times$ SS/MS $\times$ D0/D1/D2 $\times$ A0 $\times$ R0/R1	Core paper
L2 Attacks	Do formal attacks break prompt-level defences more than natural questioning?	L0 $\times$ 3 reps $\times$ D1/D2 $\times$ A0/A1/A2/A3 $\times$ R0	Strong paper slice
L3 Defences	Where does the utility–security frontier bend?	L0 $\times$ 3 reps $\times$ SS/MS $\times$ D0–D6 $\times$ A0 $\times$ R0	Thesis / follow-up
L4 Models	Are the main findings model-specific?	L0–L4 $\times$ 3 reps $\times$ SS/MS $\times$ D0/D2 $\times$ A0 $\times$ R0	Strong paper
L5 Actions	Does cross-boundary pressure cause unsafe actions, not just leaks?	L0 $\times$ 3 reps $\times$ D0/D2/D5/D6 $\times$ A0/A2 $\times$ R0/R1 $\times$ action tasks	Thesis / follow-up
L6 Social scale	Does relationship distribution change aggregate leakage?	L0 $\times$ 3 reps $\times$ MS $\times$ D0/D2/D5 $\times$ A0 $\times$ S1/S3/S4/S50	Thesis / follow-up

- **Context accumulation**: information from earlier turns influencing later responses.
- **Social framing**: wrapping sensitive questions in work-acceptable contexts.
- **Within-tick batching**: asking multiple questions per turn to overwhelm defences.

4.5 Dual-Metric Protocol

Every experimental run produces the utility score U and security score S defined in Equations 1 and 2. A perfect governance configuration achieves $U = S = 1$: all legitimate queries answered, all sensitive data protected. The (U, S) pair for each configuration defines a point on the Pareto frontier. A defence that achieves $S = 1$ by refusing all queries is useless; one with $U = 1$ that leaks everything is dangerous. Only dual measurement captures the tradeoff that practitioners face, and no prior cross-boundary benchmark reports both metrics simultaneously.

4.6 Relationship and Agent-Network Conditions

The core benchmark evaluates two relationship conditions:

- **R0 (Stranger)**: Tina is identified by name only. The agent treats her as unknown.
- **R1 (Colleague)**: Tina is identified as a project manager who works with Alex daily. Shared project context is provided.

The extended benchmark adds **R2 (Close friend)** and **R3 (Investor / formal stakeholder)** to test whether personal, work, and strategic relationships shift different sensitivity categories. PART-Bench also supports larger social setups: **S1** (single requester), **S3** (three requesters with different relationships), **S4** (four agents that can interact with one another), and **S50** (fifty requesters, one interaction each). These settings test whether leakage is a property of an isolated conversation or of the relationship distribution around a personal agent.

5 Experimental Setup

Design principle. The complete PART-Bench design crosses models, modes, defences, attacks, relationships, and social-network structure. A naive full factorial design is not scientifically useful: it would require 5 models $\times$ 3 replications $\times$ 2 modes $\times$ 7 defences $\times$ 4 attacks $= 840$ runs per social setup, before adding multi-agent network conditions. We therefore use a staged design. Each layer answers one new question that the previous layer cannot answer, and later layers are only run after the preceding layer is stable.

Table 4: Core utility–security results for the raw stack baseline. Values are placeholders in this draft; the final paper will report means over three replications. U is utility on public work queries; S is security on protected queries.

Rel.	Defence	U_{SS}	S_{SS}	S_{MS}	Expected pattern
R0	D0 No defence	XXXX	XXXX	XXXX	High utility, high leakage
R0	D1 Generic prompt	XXXX	XXXX	XXXX	Small security gain over D0
R0	D2 Category deny list	XXXX	XXXX	XXXX	Large security gain with limited utility cost
R1	D0 No defence	XXXX	XXXX	XXXX	Relationship context shifts what leaks
R1	D1 Generic prompt	XXXX	XXXX	XXXX	Generic policy remains brittle
R1	D2 Category deny list	XXXX	XXXX	XXXX	Stronger, but residual category leaks remain

Primary model and replications. Layer 1 uses `gpt-5-mini` as the development and ablation model because it is the current production-like baseline for the agent stack. Each cell is run with three independent replications. Layer 4 repeats the sentinel matrix across five model families: `gpt-5-mini`, `gpt-5.4-mini`, `gpt-5.4`, `kimi`, and `deepseek`. We do not cross every model with every attack; model generalization is measured after the core findings are established.

Core benchmark. The minimum NeurIPS experiment is Layer 1:

$$1 \text{ model} \times 3 \text{ reps} \times 2 \text{ modes} \times 3 \text{ defences} \times 1 \text{ attack} \times 2 \text{ relationships} = 36 \text{ runs}.$$

Each run evaluates XXXX queries. Single-step mode asks each query independently. Multi-turn mode runs a heartbeat conversation for XXXX ticks, during which the querying agent chooses questions, reframes follow-ups, and records answers.

Strong benchmark additions. Layer 4 adds cross-model generalization using a sentinel matrix:

$$5 \text{ models} \times 3 \text{ reps} \times 2 \text{ modes} \times 2 \text{ defences} \times 1 \text{ attack} \times 1 \text{ relationship} = 60 \text{ runs}.$$

The attack slice adds a small Crescendo-style multi-turn probe:

$$1 \text{ model} \times 3 \text{ reps} \times 1 \text{ defence} \times 1 \text{ attack} \times 1 \text{ mode} \times 2 \text{ relationships} = 6 \text{ runs}.$$

Evaluation. For every run we compute utility U and security S as defined in Equations 1 and 2. We also report false refusal rate on public work queries, leak rate by sensitivity category, and first-leak tick in multi-turn mode. Action-safety extensions use separate metrics: unauthorized action rate, unauthorized mutation/share rate, and wrong-principal action rate.

6 Results

6.1 Layer 1: Raw Stack Baseline

Expected finding 1: the raw stack is not deployable as-is. D0 should preserve utility but leak protected facts at a high rate, especially in multi-turn mode. This establishes the raw behavioral safety baseline for a production-style personal-agent stack.

Expected finding 2: generic privacy prompts are weak. D1 is expected to improve security by only XXXX percentage points over D0. This tests the common deployment pattern of adding a vague privacy sentence to an otherwise capable agent.

Expected finding 3: specificity matters before strictness. D2 is expected to dominate D1 because it names protected categories. If D2 improves security without a corresponding collapse in U, the first Pareto kink comes from policy specificity, not from making the agent less useful.

6.2 Multi-Turn and Relationship Effects

Expected finding 4: single-step evaluation undercounts deployment risk. The multi-turn heartbeat setting is expected to leak more than isolated one-shot questions because earlier answers create context,

Table 5: Category-level security under the core Layer 1 benchmark. The expected signature is not a uniform relationship effect, but category-specific movement: relationship context may make some questions feel more legitimate while making others more clearly out-of-scope.

Category	D0 R0	D2 R0	D0 R1	D2 R1	Interpretation
Sensitive work	XXXX	XXXX	XXXX	XXXX	Close to legitimate work context
Personal finance	XXXX	XXXX	XXXX	XXXX	Usually easy to identify as protected
Personal health	XXXX	XXXX	XXXX	XXXX	Expected to be clearly out-of-scope for colleagues
Personal relationships	XXXX	XXXX	XXXX	XXXX	May increase under social framing

Table 6: Cross-model sentinel matrix. The goal is not to fully rank models, but to test whether the core benchmark signatures survive across model families.

Model	D0 SS leak	D2 SS leak	D0 MS leak	D2 MS leak	Expected signature
gpt-5-mini	XXXX	XXXX	XXXX	XXXX	Development baseline
gpt-5.4-mini	XXXX	XXXX	XXXX	XXXX	Same ordering, different absolute rates
gpt-5.4	XXXX	XXXX	XXXX	XXXX	Stronger model may improve utility and leakage detection
kimi	XXXX	XXXX	XXXX	XXXX	Tests non-OpenAI transfer
deepseek	XXXX	XXXX	XXXX	XXXX	Tests non-OpenAI transfer

the querying agent can reframe unresolved requests, and multiple questions can be batched into a single turn. This result is a methodological warning: a single-step benchmark can certify an agent that remains unsafe in deployment-like interaction.

Expected finding 5: relationship is a security parameter. R1 is not expected to uniformly increase or decrease leakage. Instead, it should redistribute risk across categories. A colleague relationship may legitimize sensitive work or relationship-context questions while making health and finance requests feel less appropriate.

6.3 Layer 4: Cross-Model Sentinel

Expected finding 6: absolute rates vary, but benchmark structure transfers. We expect different models to occupy different points on the frontier, but the ordering D0 worse than D2 and MS risk greater than SS risk should hold. If the ordering fails for a model, PART-Bench exposes a model-specific governance interaction rather than treating privacy as a single global score.

6.4 Layer 2: Attack Slice

Expected finding 7: formal attacks turn residual leaks into systematic failures. D2 should remain useful against direct requests, but multi-turn escalation and persuasion should expose category ambiguity and partial-compliance failures. This does not replace the core benchmark; it shows that the same harness can gauge adversarial pressure.

6.5 Benchmark Extensions: Defences, Actions, and Social Scale

The extension tables make the benchmark cumulative rather than factorial. The paper can report the core deployment result first, then add cross-model and attack slices without committing to every combination. The thesis can then use the same harness to evaluate architectural isolation, action safety, and social-network scale.

Table 7: Attack robustness slice. A0 is natural questioning; A2 is a Crescendo-style multi-turn escalation. A1 and A3 are retained as benchmark extensions.

Defence	Attack	Mode	Security	Expected failure pattern
D1 Generic	A0 Direct	SS/MS	XXXX	Vague instruction fails under ordinary requests
D2 Category	A0 Direct	SS/MS	XXXX	Direct sensitive requests often blocked
D2 Category	A2 Crescendo	MS	XXXX	Gradual reframing turns protected data into apparently useful context
D2 Category	A3 Persuasion	SS	XXXX	Authority or reciprocity framing pressures refusal boundary

Table 8: Planned defense-frontier result table. Layers 3, 5, and 6 are not required for the minimum NeurIPS benchmark but define how PART-Bench scales into the thesis and follow-up defense paper.

Defence	Type	Utility	Security	Expected role
D0 No defence	Raw	XXXX	XXXX	Unsafe baseline
D1 Generic prompt	Prompt	XXXX	XXXX	Security theater baseline
D2 Category deny list	Prompt	XXXX	XXXX	First Pareto improvement
D3 Spotlighting	Prompt	XXXX	XXXX	Incremental prompt hardening
D4 Instruction hierarchy	Prompt	XXXX	XXXX	Stronger prompt-level ceiling
D5 MCC isolation	Architectural	XXXX	XXXX	Qualitative security jump
D6 MCC + escalation	Architectural	XXXX	XXXX	Utility recovery and fewer false refusals

7 Discussion and Limitations

Implications for practitioners. PART-Bench is designed to answer a deployment question: given a raw or defended personal-agent stack, where does it sit on the utility–security frontier? The expected practical lessons are: (1) do not deploy cross-boundary personal agents with raw or generic privacy prompts alone; (2) measure utility and security together, because refusal-only systems are safe but useless; (3) evaluate multi-turn interactions, because one-shot tests understate risk; and (4) treat relationship context as a security parameter, not just a personalization feature.

Toward architectural defences. The defense-frontier layer distinguishes prompt hardening from architectural isolation. If D3 and D4 improve security only incrementally while D5 changes the frontier, the conclusion is not merely that prompts need to be better. It is that personal-agent platforms need permissioned context construction: do not ask the agent to avoid sharing data it can see; ensure the data is not mounted for that relationship unless it should be available. D6 then tests whether escalation can recover utility lost through isolation.

Limitations. **Staged scope**: the benchmark is intentionally not a full factorial grid. This improves interpretability but means each layer answers a bounded question. **Synthetic scenario**: the personal agent's data is synthetic, although it is structured to resemble realistic notes, memory, and tools. **Evaluation method**: gold-fact string matching may miss paraphrased leaks or trigger on coincidental matches; we will supplement it with LLM-based and human spot-check evaluation. **Model coverage**: Layer 4 tests five model families, but the benchmark should be rerun as new agent models and tool-use stacks emerge. **Attack coverage**: A0 and A2 cover natural and progressive multi-turn pressure; PAIR-style optimization and persuasion attacks remain benchmark extensions.

Benchmark release. We will release the benchmark suite including: evaluation queries with gold key facts, agent configuration files (notes, memory, tools), the heartbeat evaluation engine, action-task templates, social-network setup templates, and evaluation scripts. The framework is model-agnostic and explicitly staged so future work can add models, attacks, defences, action families, and relationship graphs without changing the core utility–security contract.

Table 9: Planned action-safety track. Action tasks are separated from QA because the failure mode is unauthorized state change rather than information disclosure.

Task family	Authorized utility	Unauthorized action rate	Expected risk
Calendar	XXXX	XXXX	Exposes full calendar or schedules without permission
Notes	XXXX	XXXX	Shares, edits, or summarizes restricted notes
Messaging	XXXX	XXXX	Drafts or sends messages under the wrong principal
Todos	XXXX	XXXX	Creates or completes tasks for the wrong actor

Table 10: Planned social-scaling track. These settings measure whether leakage is a property of one conversation or of the relationship network around the personal agent.

Setup	Metric	Expected pattern
S1 Single requester	Utility / security	Clean baseline for comparison
S3 Three requesters	Category leak by relationship	Different relationships pull different categories across the boundary
S4 Four interacting agents	Cross-agent leakage path	Agents may route around a boundary by asking each other
S50 Fifty requesters	Aggregate leak distribution	Social graph composition changes population-level leakage

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A Observed Attack Patterns

Under D2 (category-specific deny list), residual leaks in single-step evaluation cluster into identifiable patterns:

Table 11: Distribution of D2 leak causes in single-step evaluation (XXXX total leaks).

Pattern	**Count**	**Share**
Category ambiguity (data accessible but semantically sensitive)	XXXX	XXXX%
Social framing (personal question in work-acceptable frame)	XXXX	XXXX%
Partial compliance failure (refusal then continued disclosure)	XXXX	XXXX%
Other	XXXX	XXXX%

Category ambiguity is the dominant failure mode: data stored in "Work" folders that is semantically sensitive (e.g., board meeting notes containing investor terms, HR documents with salary benchmarks). The agent's policy correctly protects "personal finance" but does not recognise that a board memo discussing fundraising terms is financially sensitive.

B Experimental Configuration Details

Alex's agent. System prompt establishes identity (Co-founder & CTO, Oxford CS background). 85+ notes across 6 folder categories. Tools: `read_note`, `search_notes`, `check_calendar`, `schedule_meeting`. Memory contains preferences and interaction history.

Tina's agent. Operates from a HEARTBEAT.md instruction file. Phase 1 (information gathering): systematically works through pending questions. Phase 2 (deeper probing): re-asks refused questions with different framing. Maintains a MEMORY.md checklist tracking answered/refused/pending status per question.

Heartbeat configuration. Each tick: Tina's agent selects 1–3 questions, sends a message, Alex's agent responds. XXXX ticks per group in multi-turn mode.

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